BIZTECTIVE (ROOT 2.0) · PRODUCTION ARCHITECTURE & DEPLOYMENT

Deterministic Root-Cause Investigation & Verified Recovery on Decentralized Cloud

"Every financial metric is computed deterministically. Every claim cites its source rows. Every external action is owner-approved and verified by read-back."

LIVE WEB APPLICATION

https://root-web-app.alternatefutures.ai

Service: root-web · DSEQ 29100447

LIVE FASTAPI + LANGGRAPH API

https://root-api-app.alternatefutures.ai

Service: root-api · DSEQ 29100292

VERIFIED TEST & SECURITY GATES

786 Pytest · 57 Vitest · 17 E2E

96% Backend Cov · WCAG 2.2 AA

Dashboards Show What Changed. Raw LLMs Hallucinate Why.

When a commerce business sees net margin drop across 500,000 order lines, owners need a provable root cause and a safe recovery plan — not unverified prose or blind API mutations.

FAILURE MODE 01 — Contract R-10

Invented Financial Math

Prompting an LLM over raw CSVs produces fabricated revenue deltas, miscounted restockable COGS on returns, and non-reproducible numbers across runs.

Fix: LLMs never compute numbers. All KPIs live in deterministic Python/SQL modules.

FAILURE MODE 02 — Contract R-11 / R-12

Zero Evidence Provenance

Black-box text summaries mix confirmed dataset facts with model hypotheses and user guesses, without citing dataset IDs, date windows, or row counts.

Fix: Every claim binds to an immutable `EvidencePayload` and passes a deterministic critic.

FAILURE MODE 03 — Contract R-13

Uncontrolled Side Effects

Giving an LLM agent direct write tools risks unapproved customer contact, spreadsheet formula injection, or duplicate rows on network retries.

Fix: Models have zero write tools. Execution requires owner approval of a SHA-256 plan hash.

Biztective (ROOT 2.0) Design Principle: Separate deterministic facts (`Fact`), evidence-backed root causes (`Hypothesis`), owner context (`Human Input`), bounded recovery ranges (`Estimate`), and verified writes (`Action`).

Zero Hallucinated Facts

Strict Separation of Calculation, Reasoning & Execution

How Biztective Turns Raw CSVs into Verified Operational Recovery

LAYER 1 · INGESTION

Hygiene & Quarantine

- Streaming CSV parser (≤50 MB, ≤500k rows across `orders`, `products`, `customers`, `returns`).
- Quarantines malformed/conflicting rows (`V-001..V-022`) with exact line provenance.
- Strips PII columns (`SEC-08`) and neutralizes `= + - @` formula prefixes (`SEC-09`).

LAYER 2 · ANALYTICS

Deterministic Engine

- Pure SQL + Python `Decimal` financial formulas (`app/analytics/`).
- Computes gross/net revenue, COGS, restockable return reversals, and RFM segments.
- `analysis_config v1` flags statistically significant return & margin anomalies.

LAYER 3 · LANGGRAPH

Investigation & Critic

- 10-node state machine persisted in PostgreSQL (`langgraph.checkpoints`).
- Read-only SQL evidence tools; deterministic critic (`critic.py`) rejects unsupported claims.
- Placeholder contract (`{{metric:...}}`) prevents LLM number fabrication.

LAYER 4 · RECOVERY

HITL & Sheets Executor

- Immutable versioned recovery plans (`v1, v2...`) with `sha256` content hashes.
- Owner-only approval (`recovery:approve`) + explicit execution (`action:execute`).
- Writes `ROOT Review Tasks` & `ROOT Service Queue` and verifies cell-by-cell.

```
# Deterministic Placeholder & Action Identity Contract (services/api/app/)
Action ID       = uuid5(ACTION_NAMESPACE, f"{plan_id}:{version}:{content_hash}:sheets.write_recovery_tasks")
Idempotency Key = sha256(f"{action_id}:{payload_hash}")  # Every spreadsheet row keyed by <action_id_hex>-<n> in Column A
```

Ten-Stage Verified Pipeline Across Human & Container Restarts

Investigations pause cleanly at `awaiting_human` in PostgreSQL and resume under per-thread transaction advisory locks without holding open HTTP connections or process memory.

INVESTIGATION SUBGRAPH (10 NODES)

Evidence Gathering → Critic → Clarification

- `load_context → plan_checks → run_verification`: Executes deterministic SQL checks for return spikes, vendor defect clusters, and margin drops.
- `critic → decide_next`: Verifies sample size, time coverage, and non-contradiction before admitting findings.
- `ask_human (HITL) → incorporate_human`: Pauses for up to 3 targeted owner questions; creates a new immutable `interpretation_versions` row (`v1 → v2`) on every answer.

RECOVERY & GOOGLE SHEETS EXECUTOR

Bounded Estimation → Approval → Read-Back

- `recovery/estimator.py`: Computes low/mid/high recovery ranges with explicit assumptions and caveats.
- `Decision Panel`: Owner approves, requests revision, or rejects a specific `(plan_id, version, content_hash)`.
- `actions/executor.py`: Reconciles existing sheet keys, appends missing rows, and reads back every cell (`pending → succeeded → verified`).

LIVE GOOGLE SHEETS VERIFICATION PROOF (AMD-19 / AMD-20)

Action `668e72cb-df08-5545-97d6-96e03454f2ed` wrote 1 strategy row to `ROOT Review Tasks` and 3 targeted customer rows to `ROOT Service Queue`, completing independent cell-by-cell read-back in `33.46 s`.

Status: verified

Project: root-2 (cmvbvc76110gwbs1uxtk23lhr)

Containerized Multi-Service Deployment on Decentralized Compute

Both frontend and backend run as non-root (`UID 10001`) OCI containers on **Clouds by Alternate Futures** backed by **Akash** compute providers and connected to managed PostgreSQL over TLS (`sslmode=require`).

Component	Service ID & Akash Lease	Public HTTPS Endpoint	Runtime Image & Specs	Health / Security

Frontend (`root-web`)
Next.js 16.4.2 Standalone
	`cmvbx1b1016uebs1u7l6ge6p7`
	`DSEQ: 29100447`
		https://root-web-app.alternatefutures.ai	`node:24.14.1-alpine3.21`
			`ttl.sh/root2-mahateer-web-1.0.2:24h`
			`1 vCPU · 2 GiB RAM · Port 3000`
				`UID 10001 (rootapp)`
				`CSP Nonce + X-Frame DENY`
				`GET /login → 200 OK`

Backend (`root-api`)
FastAPI + LangGraph + Worker
	`cmvbwfap014pibs1u7pxz2e5k`
	`DSEQ: 29100292`
		https://root-api-app.alternatefutures.ai	`python:3.13-slim-bookworm`
			`ttl.sh/root2-mahateer-api-1.0.2:24h`
			`1 vCPU · 1 GiB RAM · Port 8000`
				`UID 10001 (rootapp)`
				`--workers 1 (SEC-20 + Lease)`
				`GET /api/v1/readyz → ready`

Database & Checkpoints
Managed PostgreSQL (Supabase)
	`ap-southeast-2`
	`TLS sslmode=require`
		`Internal Pooler :5432`
		`Alembic Head: 0007`
			`Runtime: root_app (NOBYPASSRLS)`
			`Migrations: postgres (owner)`
			`Schema: public + langgraph`
				`22 Tenant Tables RLS Verified on every /readyz`
				`Append-only audit_log`

Why Single-Worker Uvicorn (`--workers 1`) in `root-api`

Preserves the single-instance contract (`ARCHITECTURE.md §15`): the embedded PostgreSQL job worker (`WORKER_ENABLED=true`) and the in-process `SEC-20` authenticated session token bucket (`300 req/min`) remain authoritative without external Redis or split-brain state.

Startup Migration & Least-Privilege Separation (`AMD-3`)

`docker-entrypoint.sh` runs `alembic upgrade head` via `DATABASE_URL_MIGRATIONS` (table owner), then drops privileges so FastAPI and the job worker connect exclusively via `DATABASE_URL` (`root_app` role with RLS enforced).

Solving Build-Time Next.js Rewrites for Portable Cloud Containers

Next.js evaluates `next.config.ts` `rewrites()` at `next build` time. On Clouds, service URLs are assigned when deployed — so we engineered request-time proxy rewriting in `apps/web/src/proxy.ts`.

REQUEST-TIME PROXY (apps/web/src/proxy.ts)

Dynamic BACKEND_URL Resolution

- Intercepts `/api/v1/:path*` at request time inside the Next.js server runtime.
- Reads `process.env.BACKEND_URL` injected via `acc services env set` without rebuilding the image.
- Generates per-request cryptographic CSP nonces (`script-src 'self' 'nonce-...'`) on all HTML routes.

FIRST-PARTY HTTPS COOKIE CONTRACT

__Host- Cookies + Signed Double-Submit CSRF

- Browser only talks to `https://root-web-app.alternatefutures.ai` — zero third-party cookie blocking.
- `__Host-root_session` (`Secure; HttpOnly; SameSite=Lax; Path=/`) and `__Host-root_csrf` (`SameSite=Strict`).
- Backend `CsrfMiddleware` validates `Origin == FRONTEND_ORIGIN` + HMAC-SHA256 session-bound token.

```
// apps/web/src/proxy.ts — Request-time /api/v1/* rewrite preserving portable container builds (AMD-22)
if (pathname.startsWith("/api/v1/")) {
  const target = new URL(`${pathname}${request.nextUrl.search}`, backendOrigin());
  return NextResponse.rewrite(target); // Proxies to https://root-api-app.alternatefutures.ai/api/v1/*
}
```

16 Material Findings Remediated & Regression-Locked

Hardened Against Concurrency Races, Stale Workers & Ambiguous Writes

F10-01 · F10-02 — Auth & Quotas

Serialized Login Throttling & Session Buckets

Ordered PostgreSQL transaction advisory locks serialize concurrent login attempts (12 parallel bad logins → exactly 5 `401`s and 7 `429`s; Argon2 offloaded from the event loop). Added `300 req/min` session token bucket.

F10-05 · F10-06 — Job Queue

Attempt-Fenced Worker Leases

Every job heartbeat, completion, and failure update is fenced by `(job_id, status='running', attempts=leased_attempt)`. A stale worker replaced after lease expiry cannot overwrite the new worker or fire duplicate failure hooks.

F10-09 · F10-10 · F10-11 — External Actions

Pre-Mutation Recheck & Crash Reconciliation

Approval & integration binding are re-verified in PostgreSQL immediately before every external HTTP mutation (including tab creation). Post-redirect Apps Script errors are marked ambiguous and reconciled via read-back before any retry.

Recursive Log Redaction (F10-03)

Structured JSON logger recursively redacts nested secret keys, strips SQLAlchemy bound parameters (`hide_parameters=True`), and logs only exception class + frame locations so credentials never leak to container stdout.

500k-Row Window-Function Deduplication (F10-16)

Replaced `NOT IN` retained-ID materialization with single-pass `row_number() OVER (PARTITION BY key ORDER BY source_row)`, cutting 500,000-line upload + ingestion from `>146 s` to `50.37 s`.

Zero Skipped Gates · Zero Unverified Claims

50.37 s
500K ROWS + 25K RETURNS INGEST
18.7 MB orders CSV; NFR-02 target ≤ 120 s. 100k rows in 15.02 s (target ≤ 30 s).

265 ms
P95 CONCURRENT API READS
60 authenticated requests across 20 concurrent users (101.7 req/s); target ≤ 300 ms.

96.0%
BACKEND COVERAGE (786 TESTS)
All 7 critical modules ≥ 95% (auth 99%, critic 98%, verification 97%, analytics 96%).

17 / 17
PLAYWRIGHT E2E + AXE WCAG 2.2 AA
Full workflow, keyboard-only, mobile viewport, cross-tenant 404, and 57 Vitest unit tests.

Benchmark / Verification Suite	Dataset / Scope	Measured Result (2026-10-17)	Contract Gate
500k Analysis & Anomaly Detection	500,000 orders + 25,000 returns (defect)	16.12 s (46 SQL statements, 35 flags)	≤ 60 s (Met)
Investigation Start → First Question	5 planted incidents, 17 evidence tool calls	1.202–1.570 s (deterministic) / 1.070–2.076 s (LLM)	≤ 60 s (Met)
Frontend Production Route Budgets	6 primary Next.js 16 App Router routes	TTFB 33–88 ms · Heading 275–591 ms · JS 152–273 KB	≤ 400 KB JS (Met)
Static & Security Audits	mypy --strict · ruff · tsc · pip-audit · gitleaks	0 type/lint errors · 0 prod CVEs · 0 leaked secrets	100% Clean (Met)

Redeploy & Rollback Workflow

Clouds caches image tags on provider nodes and rejects mutable `:latest` or `:main` tags. Every update bumps the version tag in `.github/workflows/deploy-images.yml` and deploys via `acc`:

```
# 1. Deploy new versioned frontend & backend images on Clouds
acc services deploy cmvbx1b1016uebs1u7l6ge6p7 \
  --image ttl.sh/root2-mahateer-web-1.0.3:24h -y
acc services deploy cmvbwfap014pibs1u7pxz2e5k \
  --image ttl.sh/root2-mahateer-api-1.0.3:24h -y

# 2. Inspect structured JSON container logs
acc services logs cmvbwfap014pibs1u7pxz2e5k

# 3. Instant rollback to previous known-good tag
acc services deploy cmvbwfap014pibs1u7pxz2e5k \
  --image ttl.sh/root2-mahateer-api-1.0.2:24h -y
```

TRANSPARENCY · KNOWN LIMITATIONS

AF Brand Voice

What Is Scoped & Bounded Today

- **Ephemeral Registry (`ttl.sh`):** Image tags expire after 24h; trigger `deploy-images.yml` before redeploying after 24 hours.
- **Single-Instance Process State:** SEC-20 session rate-limit buckets live in process memory (`--workers 1`) and reset on container restart.
- **Apps Script Bridge Latency:** Google Apps Script redirects take 2–19 s per call; bounded by 45 s timeouts and read-back reconciliation (AMD-20).
- **Deferred Baseline Items (AMD-15/17/18):** Data overlays, Slack webhooks, and self-registration UI remain out of scope by design.

Safe Schema Rollback Contract (R-14)

Container image rollbacks do not auto-downgrade PostgreSQL. All Alembic migrations follow **expand → migrate → contract** so the previous container tag stays compatible with the live schema.

Alternate Futures

Live & Verified on Clouds by Alternate Futures

09 · VERIFY THE LIVE DEPLOYMENT NOW

Try Biztective Live on Clouds by Alternate Futures

"Infrastructure that flows, not locks — paired with financial root-cause analysis you can audit down to the row."

```
# 1. Verify live production readiness & PostgreSQL RLS role invariants
curl -sS https://root-web-app.alternatefutures.ai/api/v1/readyz
{"status":"ready"}

# 2. Run the end-to-end Playwright browser suite against the public Clouds URL
PLAYWRIGHT_REMOTE_URL=https://root-web-app.alternatefutures.ai npx playwright test
```

PUBLIC WEB URL

https://root-web-app.alternatefutures.ai/login

SEEDED DEMO ACCOUNT

demo.owner@root-demo.test (Northstar Apparel)

OPERATIONAL RUNBOOK

docs/DEPLOYMENT_GUIDE.md · AMD-22